

# Content validity of the modified EQ-HWB-9 in a sample of Argentinean patients, informal caregivers, and members of the general public

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**Title:**

Content validity of the modified EQ-HWB-9 in a sample of Argentinean patients, informal caregivers, and members of the general public.

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**Abstract****Background:**

In response to the challenge of measuring the impact of health, social care, and informal caregiving on quality of life, the EuroQol Group developed the Experimental EuroQol Health and Wellbeing (EQ-HWB-9). Evidence from preliminary testing informed a series of revisions to improve the instrument's performance. This study aimed to assess the content validity of the Argentina (Spanish) EQ-HWB-9 Experimental Modified Paper Self-Complete v1.2 (modified EQ-HWB-9), to explore whether its modifications represented an improvement over the earlier experimental version, and to examine how participants ranked the frequency scale response options.

**Methods:**

Qualitative cognitive interviews were conducted using a semi-structured guide with open-ended questions and targeted probes to assess comprehensibility, relevance, and comprehensiveness of the modified EQ-HWB-9. Participants were recruited

through purposive sampling via patient associations, retirement centers, and social media. Thematic content analysis was guided by an *a priori* framework. Participants also completed a response option ordering exercise to compare their rankings with the instrument's frequency scale.

### **Results:**

Twenty participants were included: patients (n=12), informal carers (n=4), and members of the general public (n=4); nine were also social care users. The median age was 64.5 years (range 25–88) and 12 were women. All participants found the questionnaire easy to understand, items were deemed relevant, and response options appropriate. Some participants noted missing content related to leisure, social relationships, financial situation, and environment, and suggested minor wording revisions. The modified EQ-HWB-9 was generally preferred over the earlier version, supporting better comprehension and alignment with local language and context. In the frequency scale task, "none of the time" and "most or all of the time" were consistently placed at the expected extremes (90% and 85%, respectively), while intermediate labels showed greater variability, with overall ordering showing good alignment with the instrument's scale.

### **Conclusion:**

This study, conducted as part of an international multi-language effort, represents the first content validity assessment of the modified EQ-HWB-9 in Spanish and Latin America. Findings support the instrument's utility, though further refinements to item wording are recommended. Ongoing iterative testing will help ensure the tool reliably measures health and wellbeing to inform policy and practice.

### **Key words**

EuroQol Health and Wellbeing 9 (EQ-HWB-9); Content validity; Cognitive interviews; Health and wellbeing measurement; Argentina

### **1. Background:**

Health-related quality of life (HRQoL) is a key outcome in the evaluation of health interventions, as it reflects an individual's subjective perception of their physical, emotional, and functional status in clinical and social contexts. Economic evaluations commonly use HRQoL as part of quality-adjusted life years (QALYs), a composite measure combining survival and quality of life, to assess treatments and health technologies.(1–3) Traditionally, instruments used to measure HRQoL have either been designed for use in populations experiencing specific health conditions, or they have been designed for use in a wide range of populations (the so-called

'generic' measures). Fewer measures have focused on populations indirectly affected by illness, such as informal caregivers, family members, or users of social care services. It has been suggested that generic preference-based measures (GPBMs) of HRQoL, while widely used in health technology assessments, may not capture all domains relevant to informal caregivers, social care users, and the general population, including caregiver burden, reduced autonomy, or a lack of social recognition (4).

In response to these considerations, the EuroQol Group developed the EQ Health and Well-being (EQ-HWB) questionnaire, a generic measure that captures both health and broader aspects of wellbeing, enabling its use across health, social care, and community settings(5). Two experimental versions of EQ-HWB were developed: a long form with 25 items and a short form with 9 items (EQ-HWB-9), the latter designed to enable the estimation of QALYs. Their development was grounded in an extensive review and involved testing across multiple populations and languages, including Argentine Spanish.(6-8). Following initial testing and early applications, further evaluations led to refinements resulting in a experimental modified version of the EQ-HWB-9, which is the instrument examined in the present study.

A critical aspect in the validation process of any such instrument is assessing content validity, defined as the degree to which the instrument's items adequately reflect the construct intended to be measured within the target population(9). Content validity is especially important in cross-cultural contexts, where the meaning attributed to health, wellbeing, and quality of life may vary considerably. International guidelines, such as those provided by COSMIN, recommend assessing content validity for each language version and population in which the instrument is intended to be used, ensuring clarity, relevance, comprehensiveness, and semantic equivalence with the original version.(10) In line with these recommendations, the EuroQol Group promoted the conduct of content validity studies of the EQ-HWB-9 across several countries and languages.

This study aimed to assess the content validity of the Argentina (Spanish) EQ-HWB-9 Experimental Modified Paper Self-Complete v1.2 (hereafter referred to as the modified EQ-HWB-9) among patients, social care users, and informal carers. The study also explored whether the modifications introduced in the modified EQ-HWB-

9 represented an improvement over the earlier version, the Argentina (Spanish) EQ-HWB-9 Experimental Paper Self-Complete v1.1 (hereafter referred to as the EQ-HWB-9). A secondary aim was to examine how participants ranked the response options of the frequency scale proposed in the modified EQ-HWB-9.

## 2 Methods

This study, conducted between October and December 2024, primarily used qualitative methods, with an additional response option ordering task based on a card-sorting exercise. The qualitative phase consisted of semi-structured interviews, using cognitive debriefing techniques to assess comprehensibility, relevance, and comprehensiveness. Because content validity requires subjective input from the target population qualitative approaches are particularly appropriate for this purpose.(9,11) The reporting of this manuscript was guided by the COREQ (Consolidated Criteria for Reporting Qualitative Research) checklist.(12) The response option ordering exercise, focused on examining how participants ranked the response options of the frequency scale included in the questionnaire.

### 2.1 Description of the measure

The modified EQ-HWB-9 is a short-form health and wellbeing measure developed by the EuroQol Group(5). It comprises nine items covering the following dimensions: usual activities, mobility, energy, loneliness, cognition, anxiety, sadness, control over life, and physical pain. Items are phrased to capture the respondent's experience over a specified recall period.

Response options are presented using three types of five-level scales, depending on the construct being measured: a difficulty-based scale ranging from *no difficulty* to *unable*; a frequency-based scale ranging from *none of the time* to *most or all of the time*; and a severity-based scale ranging from *no pain* to *very severe pain*.

The modified EQ-HWB-9 represents a refined version of the original EQ-HWB-9, incorporating wording changes to three items (mobility, cognition, control), layout and order changes to two items (mobility, activities), and wording changes to two of the three response scale types (difficulty and frequency). (*See Supplementary Material: Table detailing the modifications from the original version.*)

### 2.2 Translation

A translation of the EQ-HWB-9 and the modified EQ-HWB-9 from the original UK

English version into Argentinian Spanish was conducted in accordance with international translation guidelines. (13) As translations of the candidate items included in earlier development phases of the instrument were already available in Spanish for Argentina (6), the forward translation from the UK source version drew on those previous translated versions, adapting them to reflect the modifications in the current versions. The translation process also included one backward translation conducted by a professional translator with no prior knowledge of the EQ-HWB versions; reconciliation of the two versions; review and approval by an expert from the EuroQol Foundation Version Management Committee; and final approval by the study team.

### **2.3 Sample**

A purposive theoretical sample of 20 participants was determined to be appropriate for this study based on guidelines for qualitative questionnaire validation(9). The sample included general public (i.e., healthy individuals), people with chronic conditions (e.g., diabetes, hypertension, asthma, or long-lasting diseases such as cancer), as well as age-related conditions (e.g., frailty, arthritis, or cognitive decline), users of social care services (individuals requiring assistance with daily living activities due to age, disability, illness, or other social circumstances, including personal care, household tasks, or emotional support), and informal caregivers (defined as individuals providing unpaid care or support to a family member, partner, or friend with a health condition, disability, or age-related needs). All participants were adults (18+years), Argentinian, native Spanish speakers, and resided in the city of Buenos Aires. Efforts were made to ensure representation across different age groups, educational levels, and genders.

### **2.4 Recruitment and consent**

Participants were recruited through multiple channels: cancer patient associations that shared the study invitation via social media and WhatsApp groups; the social media platforms of the Institute for Clinical Effectiveness and Health Policy (IECS); outreach at a senior center, where staff facilitated contact with interested members; and the research team's informal networks. The research team then contacted potential participants to schedule interviews at a mutually convenient date and time. Participants were provided with an information sheet, which they read prior to the interview. Any questions were addressed by the interviewer, and verbal consent to participate was obtained before proceeding with the interview.

## 2.5 Procedures

Cognitive debriefing was used to assess participants' understanding and ability to respond to the modified EQ-HWB-9 using both think-aloud and retrospective probing techniques. A semi-structured topic guide was used, developed in accordance with EuroQol guidelines and informed by previous content validity studies.<sup>(9)</sup> Participants were asked to review the questionnaire instructions and items and to explain their understanding in their own words. Probing focused on clarity of wording, interpretation of key concepts, appropriateness of examples, and ability to map responses to the available response options. Comparative questions assessed differences between modified and experimental versions of items (e.g., wording changes, response options, and item order), as well as participant preferences (9,14). The relevance of each item was explored in relation to participants' health or caregiving role; and overall comprehensiveness was assessed by asking participants to identify missing concepts and redundancies. The interview guide was piloted in the first two interviews, which were conducted jointly by both interviewers. The full cognitive interview topic guide is provided as supplementary material.

Interviews were conducted in Spanish by one of two interviewers (NI and FArg), with prior experience in cognitive debriefing. The interviewers had no prior relationships with the participants. Interviews were carried out individually, either face-to-face (n = 18) or via Zoom videoconferencing platform (n = 2). Face-to-face interviews took place in a range of settings, depending on participants' preferences and feasibility, including participants' homes, public locations such as cafés and parks, and at the community senior center (private outdoor patio). For virtual interviews, all materials were adapted into a PowerPoint presentation shared on-screen. During the cognitive interview, the items under discussion were displayed visually.

Participants also completed a card-sorting exercise. They were given five cards, each displaying one of the frequency response options from the modified EQ-HWB-9 ("none of the time," "a little of the time," "sometimes," "often," "most or all of the time"), presented in random order. Participants were asked to arrange the cards from most frequent to least frequent. To allow comparison with the EQ-HWB-9, a sixth card labelled "Only occasionally"—a response option replaced by "A little of the time" in the modified version—was then introduced. Participants were asked to place this card

along the scale they had just created. In virtual interviews, the response options were presented in random order on a single slide, and participants verbally indicated their preferred ordering while the interviewer arranged the options accordingly and confirmed the final sequence with the participant.

All interviews were audio-recorded and transcribed verbatim in Spanish.

Transcripts were reviewed for accuracy prior to anonymization. Interviews lasted between 20 and 50 minutes. Participants received a supermarket voucher as compensation for their time.

## **2.6 Analysis**

Spanish-language transcripts were analysed using thematic content analysis guided by an *a priori* framework (15). This framework was informed by established standards for assessing content validity and included the key dimensions of comprehensiveness, relevance, and comprehensibility. To systematically compare and contrast participants' views, a data extraction matrix was developed.

Two experienced qualitative researchers (MB and NI) extracted data from the transcripts into Excel-based matrices. To enhance trustworthiness, the data from the first two interviews were independently extracted by two researchers, and the resulting interpretations were discussed to refine the analytic framework. This final framework was then applied across all transcripts. Subsequently, one researcher conducted the extraction for each interview, which was then reviewed by a second researcher. Any discrepancies were discussed and resolved through consensus.

To analyse the card-sorting exercise, the distribution of responses and their order were first described. The expected position of each response option was the compared with the observed position assigned by participants. This analysis was conducted in two ways: first, by examining the response pattern of each individual participant; and second, by analysing the response pattern associated with each card position.

## **3 Findings**

### **3.1 Participants**

A total of 20 participants were included, comprising patients (n=12), informal carers (n=4), and members of the general public (n=4). From the total, 9 (45%) were social care users, including people with disability certificates and/or affiliates of a



retirement centre. The median age was 64.5 years (range 25–88 years), and 12 (60%) participants were women. Thirteen (65%) reported having health conditions such as kidney disease, maculopathy, diabetes, hypertension, and spinal cord injury. In terms of education, half of the participants had completed tertiary or university-level studies, 7 (35%) had finished secondary education (mandatory level), and 3 (15%) had not completed high school. (Table 1).

[Table 1 about here](#)

### **3.2 Content validity**

Participants found the modified EQ-HWB-9 to be clear and easy to understand. The items were considered acceptable, with no reports of content being inappropriate or judgmental. However, in relation to comprehensiveness, they mentioned that some aspects were not captured by the instrument, as detailed below. Regarding the comparison with the EQ-HWB-9, no clear preference among participants emerged.

#### **3.2.1 Comprehension and interpretation of the items included in the Modified EQ-HWB-9**

Table 2 presents participants' interpretations of each questionnaire item. The instructions for completing the questionnaire were well understood and perceived as intuitive by participants. The use of a 7-day recall period was generally considered appropriate by the participants. In general, most items were interpreted in line with the intended meaning. While only a few items presented minor difficulties, these did not obstruct interpretation or affect participants' ability to respond. However, one item was misinterpreted by several participants.

[Table 2 about here](#)

In the case of “Daily Activities”, only one participant misunderstood this item, interpreting difficulties as related to external factors such as not finding what they were looking for while shopping, lacking money, or facing issues with payment methods. In “Mobility”, one participant found the wording confusing—particularly the examples provided in parentheses—and felt that the question assumed the respondent already had mobility issues.

Regarding “Feeling Exhausted”, all participants understood the item clearly, associating it with both physical and mental fatigue, as well as a lack of energy or motivation. A few distinguished between exhaustion and ordinary tiredness, viewing

the former as more intense or persistent. “Loneliness” was also consistently understood and considered well phrased. Participants associated it with emotional isolation, absence of support, or limited social contact, and some differentiated between being physically alone and feeling emotionally disconnected.

Regarding “Concentration”, only one participant demonstrated limited understanding. Although the participant grasped the concept, she struggled with the phrasing and was unsure whether the question referred to focusing on a specific task, as if part of the sentence were missing. The item on “Anxiety” elicited varied interpretations. While some participants associated it with a psychiatric condition, others understood it more broadly as indecisiveness or nervousness. Although “Sadness and Depression” was generally well understood, participants differentiated between the two terms used in the item wording. *Sadness* was typically interpreted as a temporary emotional state, while *depression* was understood as a mental health condition.

The item related to “Control Over Life” was the most difficult for participants to understand. Three participants misinterpreted the item: one understood it as being dominated or controlled by others; another associated it with always needing others' approval to make decisions and feeling insecure; and one explicitly stated that he did not understand the question. In some cases, participants noted that the item required rereading, and one mentioned being unfamiliar with the word “autonomy.” Overall, although the concept was challenging for some, the rest of the participants interpreted it as referring to the ability to make personal decisions, having the freedom to do what they want/need, and economic self-sufficiency.

Finally, “Pain” was universally understood. Participants referred to a range of physical experiences, from mild, localized discomfort to severe, activity-limiting pain, and several suggested that frequency, in addition to intensity, might provide a more complete assessment.

### **3.2.2 Comparison Between the Modified and Experimental EQ-HWB-9 Versions**

Overall, both versions were generally well received, but the modifications appeared

to support better comprehension and alignment with local language and context. When comparing the order of items 1 and 2, most participants reported no clear preference. Regarding the heading style, responses were mixed: six interviewees felt the experimental version (with repeated sentence stems) made the questionnaire easier to read, while seven preferred the modified version with a single introductory heading, and five participants expressed no preference. For the “Mobility” item, the difference between the wording “inside or outside” versus “inside and outside” was not specifically evaluated; therefore, no data are available to inform this modification. In the comparison of wording between “difficulties” (modified version) and “problems” (experimental version), preferences were also split. Six participants favored “problems,” five preferred “difficulties,” and one was indifferent. Several participants did not express a preference.

Regarding the control item, the term “autonomy” and changes to the wording in parentheses (included in the modified version but not in the original English version) received stronger support. Twelve participants favored the modified version, describing “autonomy” as a clear and familiar term. Three preferred the experimental version, and three were indifferent.

### **3.2.3 Relevance and Comprehensiveness**

All items were considered relevant to the construct being measured by all participants. Participants generally did not perceive any duplication or redundancy in the included items. Only one participant mentioned that the first two items (daily activities and mobility) seemed similar. However, some participants considered that some aspects related to health and well-being were missing, such as *leisure* or *use of free time*, mentioned by five participants, and whether individuals were able to enjoy their free time, mentioned by two participants

*It lacks something about free time, a question regarding well-being. We mostly talk about work-related matters, but there isn't a question asking, for example, whether you had any difficulty engaging in leisure activities. (man, 44 years old, patient with diabetes and carer)*

Although the “loneliness” item was interpreted as feeling alone, lacking emotional support, or having no social interaction with others, many participants ( $n = 6$ ) mentioned that social interaction was an important aspect that should be explicitly included in the questionnaire, as they felt it was missing in its current form. They also suggested that the ability to connect with others is

essential to avoiding loneliness.

*I think it would be appropriate to ask a question about relationships with other people, and the ability to connect with other people. At my age, it's very important not to be alone. (woman, 76 years old, patient with diabetes and hypertension)*

One participant suggested that *financial autonomy and the ability to handle expenses or emergencies* were critical for well-being.

*For many of these aspects, you need financial autonomy. For example, if you want to do something to overcome your health issue, it greatly impacts whether you can afford the medication, especially for those without health insurance. (woman, 86 years old, patient with arthritis/anemia)*

One participant proposed adding a question on *access to medication*, stating that in their experience with cancer treatment, the process of obtaining medication and dealing with related paperwork caused significant stress and anxiety, worsening their wellbeing. Another participant suggested that *external factors* affecting quality of life, such as the country's situation or public safety, could be considered. Lastly, at least three participants noted that the questions seemed somewhat negative and suggested including more positively framed questions. They proposed focusing not only on deficiencies and difficulties but also on aspects related to pleasure and enjoyment.

### **3.3 Response Option Ordering of the Frequency Scale: Card-sorting Exercise**

#### **3.3.1 Individual ranking**

The frequency response order was evaluated according to how well the observed ranking in the current exercise matched the order used in the modified EQ-HWB-9. Twelve (60%) respondents ordered responses in exactly the same way as in the questionnaire. Four (20%) participants did not place the anchor responses ("None of the time" or "Most of the time") in the expected positions. Excluding those four cases, only four participants made minor order changes, reverting the order of "A little of the time" and "Sometimes" (n=2), and "Often" and "Sometimes" (n=1), "A little of the time" and "Often" (n=1). (Supplementary Table 1).

When exploring how participants positioned the card "only occasionally", which was used in the experimental EQ-HWB-9 and replaced in the modified version by "a little of the time", we found that four participants considered it most similar to "a little

of the time.” One participant placed it between “a little of the time” and “sometimes,” and two between “a little of the time” and “often.” In contrast, eight participants paired it with “sometimes,” three with “often,” and one with “none of the time.” One participant did not provide an answer.

### **3.3.2. Response order**

Of the 100 total cards placed by the 20 subjects, 23 cards were not placed in the expected positions, with deviations of up to three positions. In most cases, there was strong agreement between the expected and selected positions (65% to 90%). Positions 1 (“None of the time”) and 5 (“All or most of the time”) showed the highest concordance with the expected position (90% and 85% respectively), while positions 2 (“A little of the time”) and 3 (“Sometimes”) showed the lowest concordance (65% and 70% respectively). (Supplementary table 1 and 2). Additional analyses are provided as supplementary material (Supplementary tables 3 and 4)

## **4 Discussion**

The EQ-HWB-9 was developed to support economic evaluation and outcomes assessment in health and social care. Given its intended use across diverse populations and settings, it is essential to ensure that the EQ-HWB-9 adequately reflects the construct of health-related well-being across different cultural and linguistic contexts; accordingly, this content validity study contributes to confirming the relevance, comprehensiveness, and interpretability of the instrument’s items in an Argentinian cohort that included patients, social care users, and informal carers. The study explored participants’ perspectives when comparing the modified EQ-HWB-9 with the EQ-HWB-9, thereby contributing to the broader, multi-country evaluation of the instrument. Notably, this is the only study within the program to test the measure in Spanish and in a Latin American context.

Overall, the items included in the modified EQ-HWB-9 were relevant to participants across all groups and were generally perceived as easy to understand, with response options adequately capturing the range of their experiences. The positive reception of the modified items suggests that the adaptations made to the modified EQ-HWB-9 enhanced its accessibility and alignment with the health and wellbeing perceptions of the local population.

While most items were well understood, several issues emerged that may guide

future refinement. For “Mobility” (“trasladarse” in Spanish), we could not conclude whether it is preferable to use “inside or outside” versus “inside and outside.” Another item warranting attention and cross-study comparison is “Anxiety” (“ansiedad” in Spanish). Although all participants found this item relevant and were able to answer it, interpretations varied, ranging from viewing anxiety as a personality trait to considering it a mental health condition. A similar pattern emerged for “Sadness and Depression” (“tristeza y depresión” in Spanish), where participants differentiated between sadness as a temporary emotional state and depression as a mental health condition, raising the possibility of response bias depending on which concept they considered when answering.

The item that presented the greatest difficulty was “Control over your day-to-day life”. In the original translation process, the term “control” was sometimes associated with negative or authoritarian connotations, or with being “out of control” in an emotional sense. The parenthetical text also caused confusion. The version tested in Argentina incorporated the term “autonomy” (“autonomía” in Spanish). While generally well interpreted, some difficulties persisted, and further evaluation is recommended.

The instrument was considered comprehensive; however, some participants identified aspects which they considered relevant to quality of life and wellbeing but which are not covered in the EQ-HWB-9. One aspect was “loneliness” (“sentirse solo/a” in Spanish), which was interpreted as feeling alone, or a lack of emotional support or social interaction. While this was considered an important, though negative dimension, several participants suggested that social interaction should be explicitly included, but as a positive dimension in the questionnaire. Other aspects mentioned as potentially missing included the ability to engage in leisure activities, economic capacity, and having a favourable environment. Some of these aspects of well-being arose during the original instrument development but were discounted from the short version of EQ-HWB during subsequent rounds. The longer version of the instrument does include additional items on social and environmental aspects, such as “feeling that people do not support you,” or “feeling unsafe,” though these are negatively worded. One item in the longer version (“feeling accepted by others”) is positively worded though it may not reflect the broader concept of social

interaction. It is difficult for any short measure of a construct such as health or well-being to capture all aspects of the construct considered important to all people. Which means there is an inherent trade-off between maintaining a brief, preference-based instrument suitable for economic evaluation and achieving greater comprehensiveness. If, as in this case for Argentina, the missing item, say 'social interaction', is considered important enough by enough respondents then one approach to dealing with the issue would be for local researchers to consider having one or two additional local dimensions available for use with the instrument in that setting. This approach avoids losing comparability with results from other countries which use the standard questionnaire, as the core dimensions are retained, but allows for country-specific exploration of relevant issues for the local population. Such additional dimensions could perhaps be scored independently of the core dimensions although, if considered important enough by local HTA bodies, could potentially be included in local valuation exercises. This approach could retain cross-country comparability while being sensitive to local issues. (16)

Another key finding relates to the evaluation of the response options in terms of the frequency scale. The response option ordering exercise proved valuable for assessing the appropriateness of the translated frequency labels in the modified EQ-HWB-9 and for comparing them with those of the experimental version. Most participants arranged the five labels in the expected order, with "none of the time" and "most or all of the time" almost always placed at the extremes. While participants experienced some difficulty when asked to rank the middle categories in isolation, these difficulties were not evident during cognitive debriefing, when the response options were presented in their intended ascending order and could be directly compared. In this context, participants reported that the scale was clear and appropriate for responding to the items, suggesting that presenting the frequency options in an ordered format supports comprehension and helps preserve the intended meaning of the labels. Importantly, these findings also have implications for valuation exercises of EQ-HWB-9 health and well-being states; viewing response options in relation to one another may provide an important interpretive reference that differs substantially from evaluating a single response statement in isolation. While the small, non-probabilistic sample and exploratory nature of the analyses preclude firm conclusions, the response ordering exercise results support the overall ordering and distinctiveness of

the frequency response options, providing useful empirical input for future refinements of the scale.

Our findings are consistent with previous multi-country content validity studies of the EQ-HWB, which have reported high overall comprehensibility of items alongside specific cultural and linguistic challenges requiring contextual adaptation (8,14,17,18). In the present study, we found that “Control over life” generated some of the widest variation in interpretation, with some participants having difficulty understanding the underlying concept. This may be due to the concept having lower relevance or frequency in Argentinian society and/or to aspects related to item translation. Indeed, it was one of the more difficult items to translate and the Argentine version uses a higher-level register term (“autonomía”) compared to the original English version. Whether or not the concept of control over life is less relevant, or less prevalent in Argentine society would require additional research, for instance, in the anthropological and/or sociological literature and research community, and that is something that we will explore moving forward. However, it should be noted that the majority of respondents considered the item relevant and were able to respond without problems. That is probably sufficient evidence to retain the item as it stands but this content validity testing has been useful in highlighting a need for further investigation of the concept in Argentina and in identifying an item which investigators need to be aware as being potentially problematic when the questionnaire is applied in settings in Argentina and which could potentially lead to systematically different responses when compared with other countries. The same is true to some extent for sadness and depression, which were viewed as different by some of the participants in the present study, though that distinction could also potentially be seen in other languages and cultures, including English, where ‘sadness’ and ‘depression’ might well not be considered equivalent. Comparing our results with those from other sites could help to clarify some of these issues, which is one of the benefits of the type of multicultural research carried out by the EQ-HWB group.

### **Strengths and limitations**

The study was conducted following the guidelines of the EuroQol Group, adhering to principles aligned with the international protocol. This enabled cross-country



comparability and allowed the Argentine data to contribute to the pool of evidence informing decisions regarding the instrument's development.

A key strength of this study is the use of sound qualitative research methods, which allowed for an in-depth exploration of participants' comprehension, interpretation, and relevance of the questionnaire items. To this end, interview guides were adapted and pilot-tested based on those developed for other cognitive interviews conducted across multiple countries, ensuring methodological consistency and comparability. The inclusion of diverse perspectives—patients, informal carers, and members of the general public—enhanced the richness of the data and contributed to identifying a broad range of potential comprehension issues. The use of native Spanish-speaking participants ensured that the findings were culturally and linguistically relevant to the target population.

The study also benefited from diversity in participants' age, gender, and educational level, including individuals with low educational attainment. Importantly, this is the first study to evaluate the content validity of the Modified EQ-HWB-9 in Spanish and in a Latin American context, providing unique and valuable insights for the wider international validation effort. Data collection and analysis were methodologically rigorous, with trained interviewers conducting the interviews and ongoing team discussions supporting the interpretation of findings. Researcher triangulation strengthened the analysis, as each transcript was coded by one analyst and reviewed by another to ensure systematic data extraction and interpretation.

The sample was restricted to Buenos Aires residents; however, consistent with qualitative research principles, the study did not aim for national generalisability. Instead, a purposive sampling strategy was used to ensure diversity in participant characteristics and experiences. Additionally, the response option ordering exercise involved a small sample size, which—although typical in qualitative research—limits the ability to draw quantitative conclusions about label ordering.

## **Conclusion**

Overall, this study provides initial empirical support for the clarity, relevance, and cultural appropriateness of the modified EQ-HWB in Spanish for Argentina. The

findings also suggest that the modifications introduced in the modified EQ-HWB represent an improvement over the earlier experimental version, supporting its use in future research and applications. By combining cognitive interviewing with a structured response option ordering exercise, it offers one of the few evaluations of this instrument in a Latin American context and the only one conducted in Spanish within the current multi-country program. The findings reinforce the measure's potential utility across health and social care settings and highlight specific areas that may benefit from refinement. Continued testing in larger and more diverse populations will be essential to further strengthen the measure's validity and to ensure its suitability for use in economic evaluation and wellbeing assessment across different cultural contexts.

## Tables

**Table 1: Sociodemographic characteristics of the Cognitive Interview participants**

|                                      | <b>N (%)</b>           |          |
|--------------------------------------|------------------------|----------|
| <b>Gender</b>                        | F                      | 12 (60%) |
|                                      | M                      | 8 (40%)  |
| <b>Age</b>                           | 18 - 44                | 6 (30%)  |
|                                      | 45 - 64                | 4 (20%)  |
|                                      | + 65                   | 10 (50%) |
| <b>Education</b>                     | Incomplete high school | 3 (15%)  |
|                                      | High school            | 7 (35%)  |
|                                      | Tertiary diploma       | 4 (20%)  |
|                                      | University             | 6 (30%)  |
| <b>Health condition</b>              | Yes                    | 12 (60%) |
|                                      | No                     | 8 (40%)  |
| <b>Users of social care services</b> | Yes                    | 9 (45%)  |
|                                      | No                     | 11 (55%) |
| <b>Carer</b>                         | Yes                    | 4 (20%)  |
|                                      | No                     | 16 (80%) |

**Table 2: Comprehension and Interpretation of Modified-HWB-9 Items**

| Component                    | Comprehension and Interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Illustrative Quote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instructions</b>          | Most participants understood the instructions as intended. They understood that they had to choose an option that describes their health and well-being over the past week.                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>"To me, it's clear. It tells me in advance that what I'm about to answer from that point on refers to my health and well-being. I understand well-being as a word that speaks about my routine or my general health during the past week." (man, 34 years old, patient with diabetes)</i>                                                                                                                                                                                                                                                                                                                                          |
| <b>Recall period</b>         | Most participants considered the 7-day recall period appropriate, as it balanced reflection and memory accuracy. However, a few participants misunderstood the timeframe or suggested alternatives, such as a shorter (24–48 hours) or longer (past month/year) period.                                                                                                                                                                                                                                                                                                                      | <i>"I think one week is fine because, well, I assume there's a reason why it's seven days and not 20 or 30 days. If it were 30 days, we might just average everything out and have the same response because something bad must have happened in the last month." (man, 44 years old, patient with diabetes and carer)</i>                                                                                                                                                                                                                                                                                                            |
| <b>Q1 - Daily Activities</b> | <p>The vast majority (19) understood the item as intended. The terms and formulation were appropriate, and examples helped clarify the question. Daily activities mentioned included working outside the home, domestic tasks, studying, shopping, errands, and recreation (sports, crafts, reading). Older participants reporting difficulty linked it to physical or emotional limitations.</p> <p><u>Interpretation difficulties:</u></p> <p>One participant misinterpreted the item, thinking it referred to difficulties like not finding what they were looking for when shopping.</p> | <p><i>"Going to work, shopping, and doing household chores seem like three totally different aspects, but having them next to each other builds the idea that it refers to all types of activities." (man, 34 years old, patient with diabetes)</i></p> <p><i>"Reading, walking, taking care of the house, taking care of grandchildren, doing some shopping, going to events." (woman, 69 years old, carer)</i></p> <p><i>"I think of someone who had problems at work, or went to the supermarket, couldn't find what they needed to buy, or didn't have enough money" (man, 44 years old, patient with diabetes and carer)</i></p> |

|                               |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q2 - Mobility</b>          | <p>Terms and formulation were mostly appropriate. Mobility was commonly understood as physical movement. For people with disabilities, mobility inside vs. outside the home differs significantly.</p> <p><u>Interpretation difficulties:</u></p> <p>One participant found the examples confusing and felt the question assumed existing mobility problems</p> | <p><i>"Mobility, activity, different physical movements, displacement." (woman 69 years old, carer)</i></p> <p><i>"I don't quite understand if it's assuming that I already have problems moving." (man, 34 years old patient with diabetes)</i></p>                                                                                                                                              |
| <b>Q3 - Feeling Exhausted</b> | <p>Well understood by all participants. The item was associated with both physical and mental fatigue: lack of energy, motivation, need to lie down. Two participants distinguished 'exhaustion' from just 'being tired.'</p>                                                                                                                                  | <p><i>"Exhaustion, general fatigue, both physical and mental..." (man, 34 years old patient with diabetes)</i></p> <p><i>"I have no desire to do anything, I'm tired." (woman, 86 years old, patient with arthrosis and anemia)</i></p> <p><i>"When I have to climb two flights of stairs, that's when I feel completely exhausted." (man, 27 years old, patient with spinal cord injury)</i></p> |
| <b>Q4 Loneliness</b>          | <p>Well understood by all participants. The terms and formulation were appropriate. It was interpreted as feeling alone, lacking emotional support, or having no social interaction with others. One participant distinguished emotional loneliness from being physically alone.</p>                                                                           | <p><i>"Someone who feels alone is missing affection, help, or someone to talk to." (woman, 74 years old, general population)</i></p> <p><i>"I can be alone in my house but feel accompanied by my memories and entertained with things to do." (woman, 86 years old, patient with arthrosis and anemia)</i></p>                                                                                   |
| <b>Q5 Concentration</b>       | <p>Well understood by most of the participants. Examples aligned with intended meaning—understanding texts, decisions. Three distinguished between 'concentration' and 'thinking clearly.'</p> <p><u>Interpretation difficulties:</u> One had limited understanding, unsure if it</p>                                                                          | <p><i>"Concentration is doing a specific task. Thinking clearly is understanding or needing help." (woman, 86 years old, patient with arthrosis and anemia)</i></p> <p><i>I don't understand if the question refers to</i></p>                                                                                                                                                                    |

|                                    |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | referred to a specific task.                                                                                                                                                                                                                                                                                                                               | <i>whether I concentrate on a particular task.<br/>(woman, 83 years old, patient with arthrosis)</i>                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Q6 - Anxiety</b>                | Well understood by most of the participants. Most linked it to an emotional state—nervousness, worry, impatience and some used the clinical concept.                                                                                                                                                                                                       | <i>"It makes me anxious trying to meet the schedule or needing help for a task and wait for the other person's response." (man, 84 years old, patient with maculopathy)</i><br><br><i>"that means being nervous" (woman, 75 years old, patient carer with hypertension and carer)</i>                                                                                                                                                                                    |
| <b>Q7 - Sadness and Depression</b> | Well understood by most participants. Only one had difficulty defining sadness. Most distinguished sadness (temporary, event-related) from depression (clinical, persistent lack of motivation).                                                                                                                                                           | <i>"Sadness seems to be linked to something specific; depression is internal." (woman, 69 years old, carer)</i><br><br><i>"Depression is an illness, sadness is a mood." (woman, 75 years old, patient with COPD)</i>                                                                                                                                                                                                                                                    |
| <b>Q8 - Control Over Life</b>      | Well understood by most of the participants. The term 'autonomy' was clearer than 'control.' Participants linked it to decision-making, economic independence, and freedom.<br><br><u>Interpretation difficulties:</u><br><br>Three had difficulty—one misunderstood the term control, one needed approvals to decide, one didn't understand the question. | <i>"Being able to decide for yourself." (woman, 25 years old, general population)</i><br><br><i>"Autonomy is when I have my freedom..." (woman, 76 years old, patient with diabetes e hypertension)</i><br><br><i>"Control... made me think of more hostile things..." (man, 34 years old, patient with diabetes)</i><br><br><i>"Those who know me, know that they can't control me and know that I am very, very determined." (woman 88, patient with hypertension)</i> |
| <b>Q9 - Pain</b>                   | Well understood by all participants. Mild pain examples: joints, head, stomach. Severe pain: fractures, trauma. Some suggested adding frequency to intensity.                                                                                                                                                                                              | <i>"Severe pain is when the pain prevents you from doing things..." (woman, 86, patient with arthrosis and anemia)</i><br><br><i>"I had very intense pain, but it was only for one</i>                                                                                                                                                                                                                                                                                   |

|  |  |                                                         |
|--|--|---------------------------------------------------------|
|  |  | <i>day." (man, 44, patient with diabetes and carer)</i> |
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## Declarations

### **Ethics approval and consent to participate:**

Protocol was approved by the local Institutional Review Board (IRB): “Comité de Ética en Investigación IRB (Iniciativa y Reflexión Bioética), Ciudad de Buenos Aires, Argentina”. On the date: September 15th, 2024. The study was conducted in accordance with the principles of the Declaration of Helsinki and applicable local ethical regulations. Verbal consent for participation in the study was obtained from all participants.

### **Consent for publication**

Not applicable

### **Availability of data and materials**

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request

### **Competing interests**

M.H., C.M. and F.A. are members of the EuroQol Group. Views expressed in the article are those of the authors and are not necessarily those of the EuroQol

Research Foundation.

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### **Authors' contributions**

MB and FAu conceived the study and developed the study materials. MB coordinated the overall project, designed the data collection and analysis plan, conducted the qualitative analysis, and led the drafting and revision of the manuscript.

CM and MH provided methodological and conceptual advice throughout the study, contributing to the development of the data collection guides and to the analytical strategy, as well as critically reviewing the manuscript for important intellectual content.

FAR and NI contributed to the study design and implementation, participated in data collection, conducted qualitative data analysis, and contributed to the preparation of the study report and manuscript.

All authors contributed to the interpretation of the findings and critically revised the manuscript. All authors approved the final version for submission, with the exception of CM, who has been unable to provide final approval due to medical leave related to health reasons. CM made substantial contributions to the study and earlier drafts of the manuscript.

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